

R&D Management Workshop 2026

Who Survives the Scientific Pivot?

Long-Term Consequences of Scientific Pivots

Saeyoung Choi Junghye Lee*

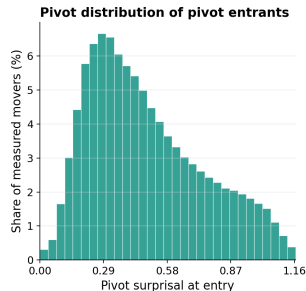
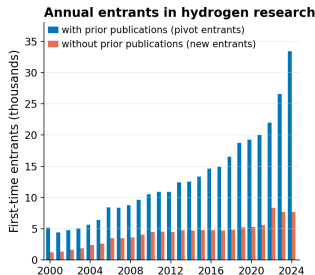
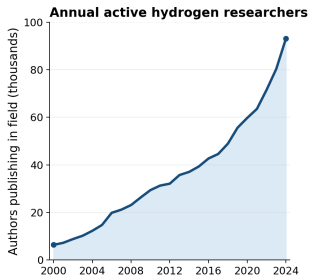
*Corresponding author: junghye@snu.ac.kr

This work was supported by the NRF Grant funded by the Korea Government (MSIT) under Grant No. RS-2025-00516776.

This work was also supported by the Ministry of Education of the Republic of Korea and the NRF of Korea (NRF-2020S1A3A2A02093277).

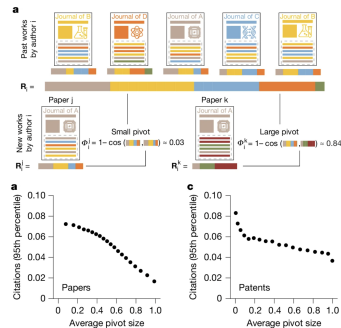
Emerging fields grow through heterogeneous researcher mobility.

- Emerging fields (e.g., hydrogen research) expanded rapidly, with increasing number of active researchers over time.
- This growth is driven especially by entrants with prior publications outside the focal field, i.e., pivot entrants.
- Among pivot entrants, their magnitude of departure from prior trajectories (pivot magnitude) varies widely.



Recent studies document the prevalence and consequences of scientific pivots.

- Research interests evolve over scientific careers (Jia et al., 2017).
- Scientists have become more likely to switch between topics over time (Zeng et al., 2019).
- Recent work formalizes the cost of pivots and documents a **pivot penalty** (Hill et al., 2025).
- Existing studies have typically focused on short-term outcomes (e.g., citation impact of the pivot paper).



Hill et al. (2025)

- Yet, little is known about what happens to scientists after they pivot to a new field, leaving the **long-term career consequences of pivots** underexplored.

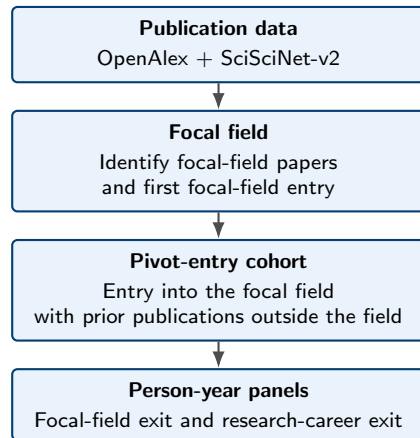
- Yet, little is known about what happens to scientists after they pivot to a new field, leaving the long-term career consequences of pivots underexplored.
- Pivot entrants differ in how far the new field is from their prior trajectory, making **pivot magnitude** a key source of heterogeneity in post-pivot outcomes.
- We examine post-pivot adaptation as a survival process: whether entrants remain in the focal field, continue their career elsewhere, or retire from research.

How does pivot magnitude shape post-pivot survival?

How does pivot magnitude shape post-pivot survival?

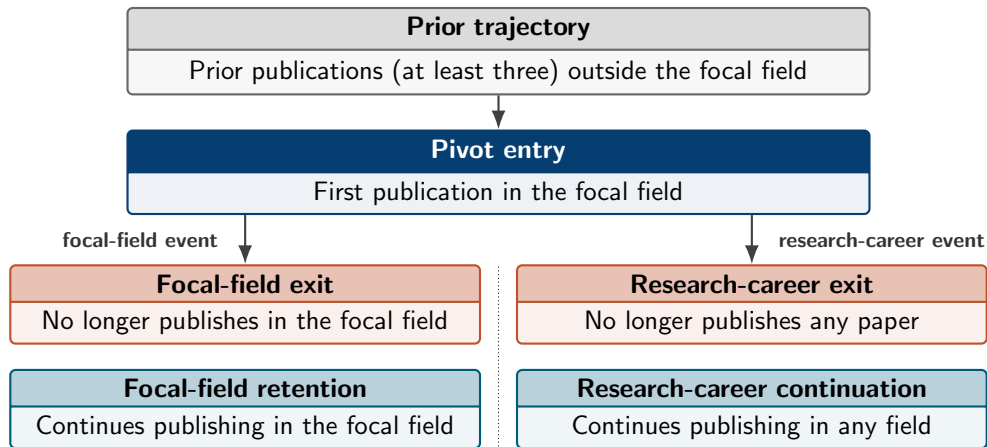
- We examine post-pivot adaptation as a survival process with two key outcomes: focal-field retention and research-career continuation.
- **RQ1:** How is pivot magnitude associated with retention in the focal field?
- **RQ2:** How is pivot magnitude associated with research-career continuation?
Does focal-field exit imply research-career exit?
- **RQ3:** Do these patterns generalize across fields?

- **Data sources:** scholarly publication data from OpenAlex and SciSciNet-v2.
- **Focal domain:** hydrogen research as an exemplary emerging field (identified by OpenAlex topics).
- **Analysis cohort:** researchers who entered the focal field between 2000 and 2021, with at least three prior non-focal-field papers.
- **Event definitions:** focal-field exit and research-career exit, measured by absence of publications in three-year windows after pivot entry.



Conceptual Framework

Separating two kinds of risk: focal-field exit vs. research-career exit



Survival and exit are measured using three-year publication windows after entry.

Discrete-time survival models estimate the risk of each event separately.

$$\Pr(\text{event}_{i,t} = 1) = f(\text{pivot magnitude}_{i,t_0} \times \text{career stage}_{i,t_0} \\ + \text{duration FE} + \text{entry cohort FE} + \text{pre-entry controls}_{i,t_0})$$

t_0 : pivot year; t : years since pivot. Standard errors clustered by author.

- **Outcome 1: focal-field exit**, i.e., three-year absence from the focal field publication.
- **Outcome 2: research-career exit**, i.e., three-year absence from any publications.
- **Pivot magnitude**: pivot surprisal (trajectory-conditioned) or pivot size (static distance).
- **Career stage**: whether the pivot entry occurs in researchers' early (1–5 years), mid (6–15 years), or late (16–30 years) career.
- To avoid post-treatment adjustment, the models condition only on pre-entry controls measured at or before the pivot year.
- Pre-entry controls: publication productivity and field diversity.

Measuring Pivot Magnitude

Predictor: pivot surprisal, a trajectory-conditioned measure of pivot magnitude.

- A sequential model predicts the next paper's references based on the prior trajectory:

$$P_{\theta}(r \in p_{t_0} \mid p_{<t_0}).$$

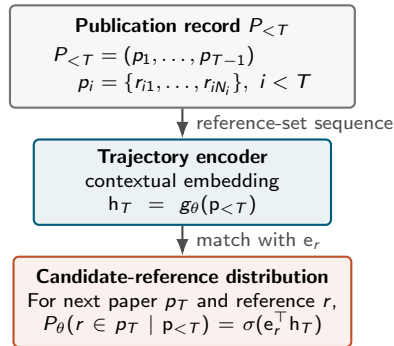
- **Pivot surprisal** measures how unexpected the pivot paper's references are:

$$S_{t_0} = \frac{1}{N_{t_0}} \sum_{r \in p_{t_0}} -\log P_{\theta}(r \in p_{t_0} \mid p_{<t_0})$$

- Larger surprisal means that the pivot paper is less expected from the prior trajectory.

- **Pivot size** (Hill et al., 2025) is used as a static-distance-based benchmark:

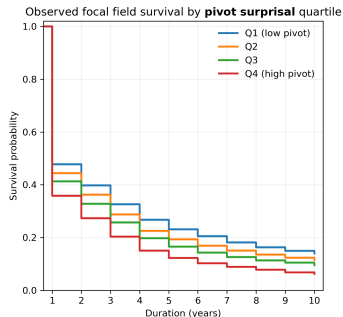
$$\Phi_{t_0} = 1 - \cos(p_{t_0}, p_{<t_0}), \text{ where } p_{t_0} \text{ and } p_{<t_0} \text{ are the aggregated reference vectors.}$$



Result 1: Focal-field Retention

Research Question 1-1

How is pivot magnitude associated with retention in the focal field?



Term	Coef.	SE	Odds Ratio
pivot surprisal (early-career, 1–5 yrs)	0.120	0.012	1.13
pivot surprisal (mid-career, 6–15 yrs)	0.210	0.008	1.23
pivot surprisal (late-career, 16–30 yrs)	0.218	0.011	1.24

Focal-field exit logit. Odds ratio is per 1 SD increase in z-standardized pivot surprisal;

$N = 200,104$ person-years (89,838 persons).

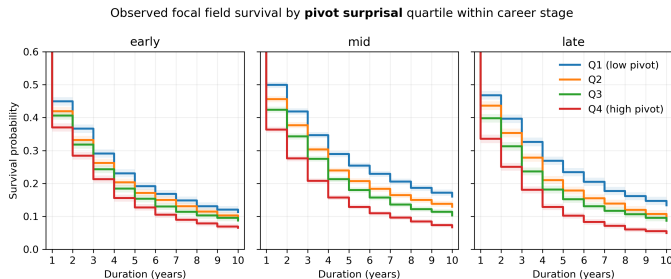
- **Larger-pivot entrants are more likely to leave the focal field.** (left, KM curve)
- Field-exit risk increases with pivot surprisal, and this pattern is consistent across career stages. (right, discrete-time logit)

Result 1: Focal-field Retention

Career-stage heterogeneity

Research Question 1-2

How do these relationships vary by career stage?

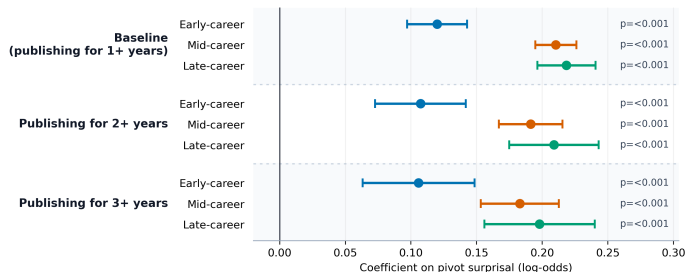


- While consistent across career stages, the pivot-survival relationship is stronger for mid- and late-career entrants than for early-career entrants.
- (smaller-pivot) Mid- and late-career entrants show relatively higher retention.
- (larger-pivot) The career stage gap narrows, as all entrants face high exit risk.

Result 1: Focal-field Retention

Robustness checks

- Baseline **pivot entry** include anyone who publishes at least one paper in the field, which may include some exploratory entries that do not reflect a true pivot.
- Re-estimate among entrants who keep publishing after the pivot year, i.e., conditional on early retention in the focal field.



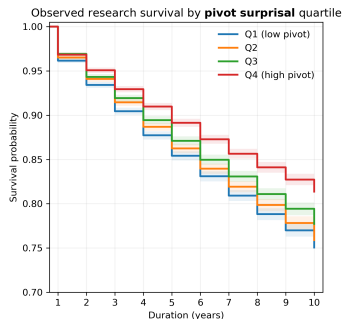
- Conditional on early retention, the field-exit association remains positive and stage heterogeneity remains consistent.

Result 2: Research-career Continuation

Research Question 2-1

How is pivot magnitude associated with research-career continuation?

Does focal-field exit imply research-career exit?



Term	Coef.	SE	Odds Ratio
pivot surprisal (early-career)	-0.166	0.016	0.85
pivot surprisal (mid-career)	-0.104	0.015	0.90
pivot surprisal (late-career)	-0.093	0.021	0.91

Research-career exit logit. Odds ratio is per 1 SD increase in z-standardized pivot surprisal;
 $N = 486,858$ person-years (89,838 persons).

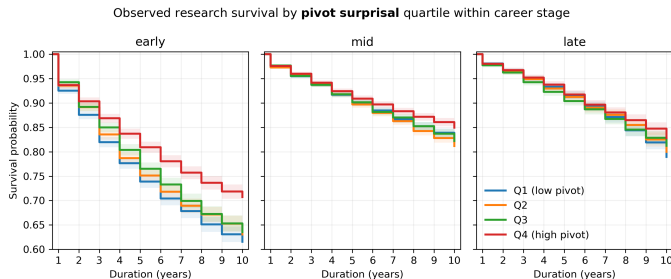
- **Larger-pivot entrants are less likely to exit research.**
- Career-exit risk decreases with increasing pivot surprisal, opposite to the field-exit pattern.

Result 2: Research-career Continuation

Career-stage heterogeneity

Research Question 2-2

How do these relationships vary by career stage?

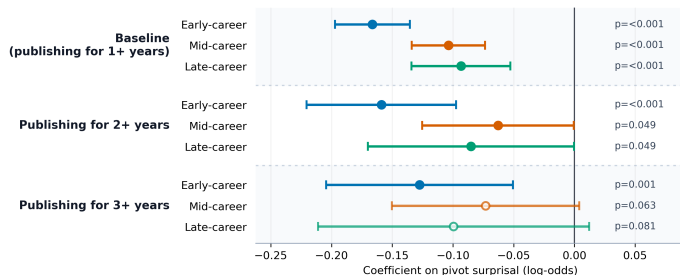


- **The pivot-survival relationship** (decreasing career-exit risk with Larger pivots) **is most pronounced among early-career entrants.**
- Nevertheless, the negative association between pivot magnitude and career-exit risk is generally consistent across career stages.

Result 2: Research-career Continuation

Robustness checks

- Re-estimate among entrants who keep publishing after the pivot year, i.e., conditional on early retention in the focal field.



- The negative career-exit association persists only among early-career entrants.
- For mid- and late-career entrants, the association is inconsistent and weaker after conditioning on early retention.

Summary of Results 1 & 2

Focal-field retention and research-career continuation show divergent relationships with pivot magnitude:

- Larger pivots are robustly associated with higher field-exit risk.
- Yet, this does not imply career-exit.
- Larger pivots do not predict higher career-exit risk (negative or null association).

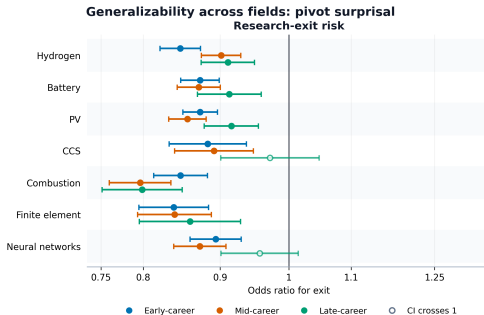
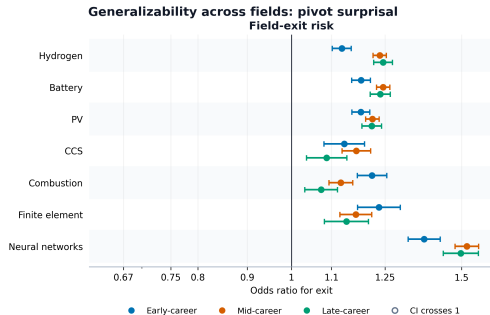
Career-stage heterogeneity is observed in both focal-field retention and research-career continuation, but the patterns differ:

- Pivot magnitude has a stronger positive association with field-exit risk for mid- and late-career entrants than for early-career entrants.
- Pivot magnitude has a consistently negative association with career-exit risk only among early-career entrants.

Result 3: Generalizability across Fields

Research Question 3

Do these patterns generalize across fields?



- Larger pivot magnitude is generally associated with higher field-exit risk across fields.
- The divergence between field-exit and career-exit patterns is common across fields.
- Career-stage heterogeneity is not consistent across fields.

Focal-field retention and research-career continuation show divergent relationships with pivot magnitude (common across fields).

→ *Why do large pivots raise field-exit risk but not career-exit risk?*

Career-stage heterogeneity is observed in both focal-field retention and research-career continuation (patterns different across fields).

→ *What drives the career-stage heterogeneity, and why does it differ across fields?*

Discussion

Distinct patterns in focal-field retention and research-career continuation

Discussion 1

Why do large pivots raise field-exit risk but not career-exit risk?

- An exploitation-exploration trade-off may underlie the divergent patterns.
- Smaller-pivot entrants may have stronger field-commitment (exploitation), as they can leverage their accumulated expertise and networks.
- Larger-pivot entrants may have weaker field-commitment (exploration), as they find it difficult to integrate their prior expertise into the focal field.
- This divergence of strategies may lead to different risk profiles: smaller-pivot entrants may be more likely to stay in the focal field, while larger-pivot entrants may put less of their career capital at risk in the focal field.

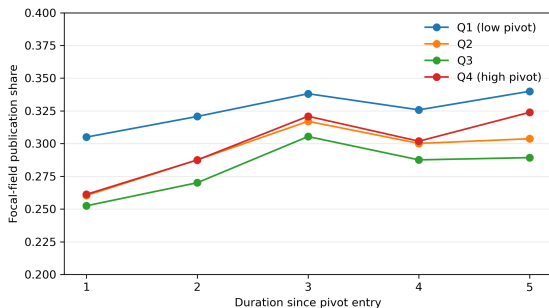
Discussion

Distinct patterns in focal-field retention and research-career continuation

Discussion 1

Why do large pivots raise field-exit risk but not career-exit risk?

- Even among researchers who stay active in the focal field, larger-pivot entrants allocate a relatively smaller share of their publications to the focal field than smaller-pivot entrants, suggesting weaker field-commitment.



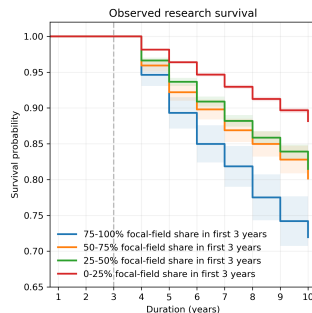
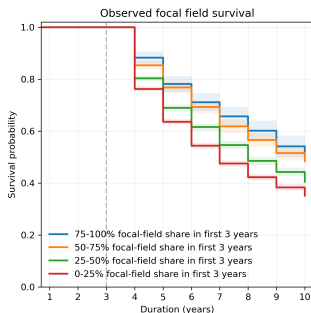
Discussion

Distinct patterns in focal-field retention and research-career continuation

Discussion 1

Why do large pivots raise field-exit risk but not career-exit risk?

- Stronger field-commitment after pivot entry (in first 3 years) is associated with higher focal-field retention and lower research-career continuation.



- While stronger field-commitment (exploitation) supports field retention, broader portfolios (exploration) help researchers survive field-level failure without exiting research.

Discussion

Career-stage heterogeneity in focal-field retention

Discussion 2

What drives the career-stage heterogeneity?

Why does it differ across fields?

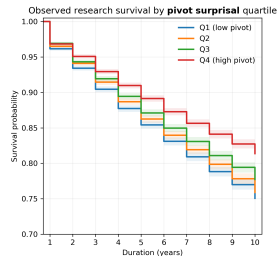
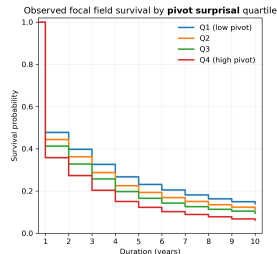
- Early-career entrants
 - (+) may be more adaptable to the new field (flexible with less sunk cost).
 - (–) have less transferable career capital.
 - (–) may be more vulnerable to failure (higher career-exit risk).
- Established entrants
 - (+) have more transferable career capital (both intellectual and social).
 - (–) may be less adaptable to the new field (rigidity with more sunk cost).
- When pivot magnitude is large, prior expertise becomes less directly transferable, weakening the career advantage in focal-field retention.
- **But this is not consistent across fields, suggesting that field conditions may shape the career-stage heterogeneity.**

Summary of findings

- Pivot magnitude is positively associated with focal-field exit risk, implying the long-term retention cost of large pivots.
- Yet, higher field-exit risk does not predict higher career-exit risk, showing divergent patterns.
- These patterns are generally consistent across fields, but career-stage heterogeneity is not, suggesting that field conditions may shape the career-stage dynamics of pivot survival.

Implications

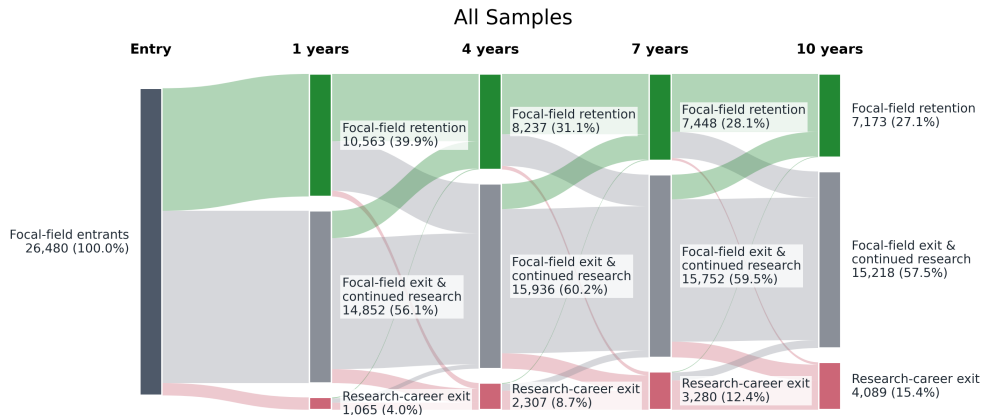
- As there are distinct patterns in long-term retention after pivot entry, we should consider the dynamics of field retention and career continuation.
- To promote researchers' adaptability to emerging fields, science policy should
 - support smaller-pivot entrants, who are more likely to stay in the focal field but face higher career-exit risk, by mitigating the career risks of early-career pivots through targeted funding and career development resources.
 - help larger-pivot entrants, who are less likely to stay in the focal field, by reducing adaptation costs through bridge funding.



Thank you for your attention!
Questions and feedback are very welcome.

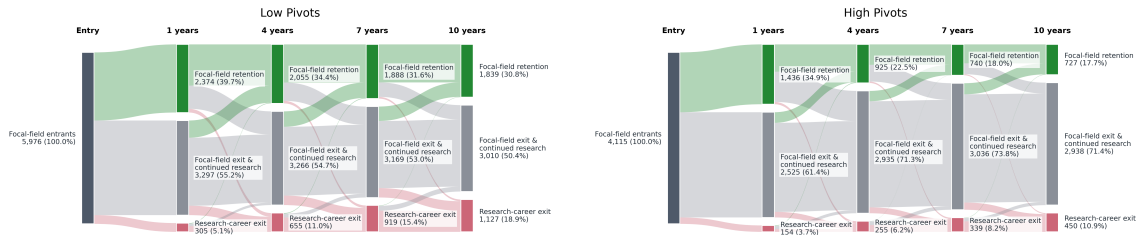
saeyoung.choi@snu.ac.kr

Backup: Sankey diagram of post-pivot trajectories



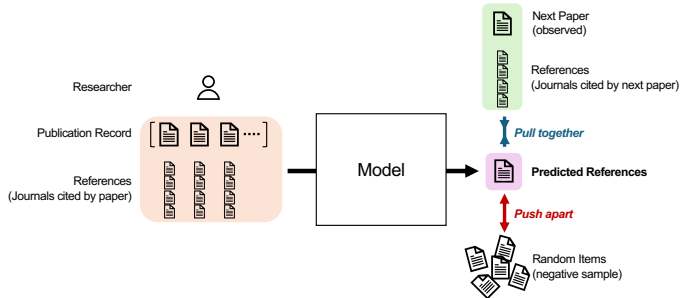
Backup: Sankey diagram of post-pivot trajectories

Although our first-event risk models cannot capture the full complexity of post-pivot trajectories, descriptive analysis of trajectory flows shows similar patterns.

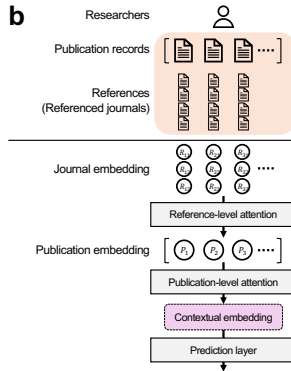


Backup: Sequential model details

a

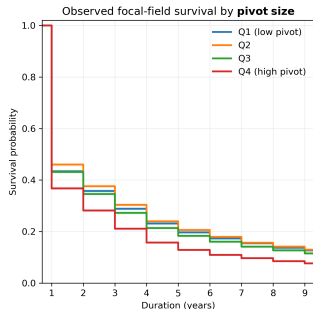
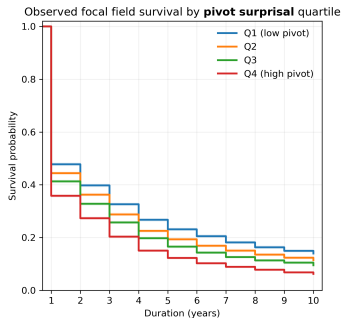


b



Backup Result 1: Metric robustness in focal-field retention

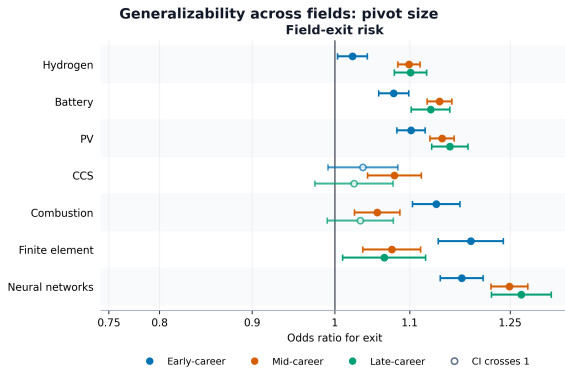
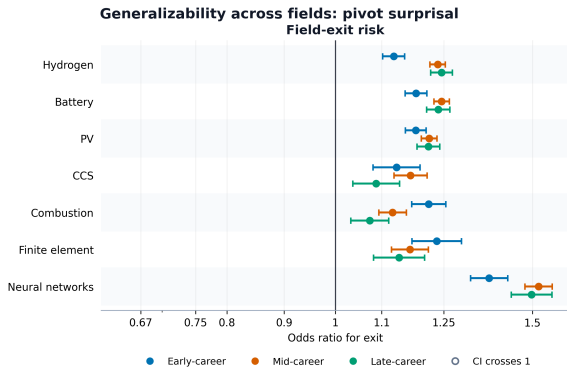
Pivot surprisal is more informative than pivot size for focal-field retention, especially among early-career entrants.



Model (pivot magnitude)	pseudo R^2	Coef.	SE	p -value
Not included (controls only)	0.1420	—	—	—
Pivot size + controls	0.1432	0.0225	0.010	.019
Pivot surprisal + controls	0.1468	0.1199	0.012	< .001

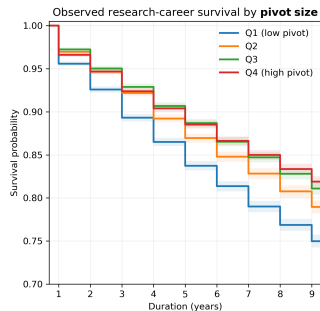
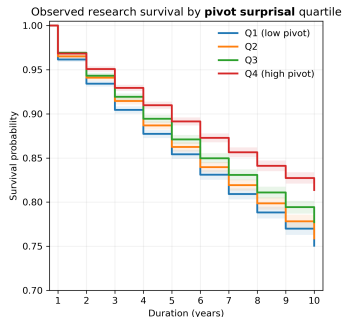
Backup Result 1: Metric robustness in focal-field retention

Pivot surprisal is more informative than pivot size for focal-field retention, especially among early-career entrants.



Backup Result 2: Metric robustness in research-career continuation

Pivot surprisal and pivot size show similar negative associations with career-exit.



Model (pivot magnitude)	pseudo R^2	Coef.	SE	p -value
Not included (controls only)	0.0482	—	—	—
Pivot size + controls	0.0556	-0.1766	0.013	< .001
Pivot surprisal + controls	0.0546	-0.1663	0.016	< .001

Backup Result 2: Metric robustness in research-career continuation

Pivot surprisal and pivot size show similar negative associations with career-exit.

